

Eco Clean – Scheduling Solution

Progress Report 5 – Upul Atapattu

1. Summary

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Project Implementation Progress

Since the last progress report, the following has been done.

Staff Management and Profile System

The staff management system was fully implemented across both backend and frontend layers. API endpoints were developed to support user creation, staff profile management, and role-based data access.

Each staff member is now associated with a structured profile, including personal details, address, and emergency contact information. The system ensures that staff can securely view and update their own information, while administrators can manage staff records centrally.

Timesheet System and Payroll Infrastructure

The payroll and timesheet system was fully developed and stabilized. A biweekly pay period structure was implemented to support payroll calculations, with automated generation of pay periods based on predefined business rules. The system integrates scheduling data with timesheet inputs, allowing staff to compare scheduled hours with actual working hours. This provides a reliable basis for calculating earnings and generating pay statements. Additionally, these fields can be overridden by the Admins manually.

All related interfaces, API endpoints, and database operations were completed and tested to ensure accurate data flow and system consistency across the payroll module.

Furthermore, the suggestive deductions based on the gross pay have been designed looking at the CRA guidelines. Since there was no free API, functions were manually created. This will help the admins to figure out the deductions before verifying it against federal and provincial systems.

Dashboard and User Interface Improvements

The admin dashboard was improved to provide a more structured and user-friendly experience. Navigation across staff management sections such as payroll, leave, and time tracking was redesigned to be clearer and more intuitive.

A responsive layout was implemented to ensure usability across different screen sizes, including mobile-friendly drawer navigation. The staff-facing PWA interfaces were also refined to improve accessibility and alignment with backend functionality. Since initial design were mainly catered to a larger screen, some of the initial UI was revamped critically.

System Integration, Testing, and Client Demonstration

Development branches were successfully merged, and integration issues were resolved to ensure a stable codebase. The system was tested using seeded data to validate end-to-end functionality across scheduling, staff management, and payroll modules. Since the staff branch was behind the main, there were few challenges when it comes to merging. However, resolving these will help us to do faster development in the upcoming week.

The web application and the PWA was demonstrated to the client, showcasing key workflows and system capabilities. Feedback was collected to guide further refinements and improvements. Furthermore, the client's hosting plans were discussed in order to actively develop manuals and documentations.

1.1. Next Steps

The following are the tasks which are planned by the next progress report deadline (23rd March 2026):

- **Optimizing and Testing the Payroll Management Flow**
- **Support Role Based Access**
- **Real life Test Data Generation**

Populate the data with the support with EcoClean to thoroughly test system functionality, validate workflows, and identify potential edge cases across the application.

- **Settings Page to Customize the solution for different companies**

Though this solution is catered to by EcoClean, we believe this solution can resolve challenges faced by many small service companies. Therefore, some effort will be made towards customizing the solution through settings.

- **Working on Final Report**

2. Worklogs

Date	Hours	Description
15th January to 16th March	96.25 hours	Previous Work
17th March 12:30 pm – 13:30 pm 8:00 pm – 9:00 pm	2 hours	Attended for a discussion with Vidarshan to understand how my staff interfaces should be linked and merged. Furthermore, discussed about tasks completed before the client meeting on Thursday.
18th March 5:30 pm – 6:30 pm 9:30 pm – 10:30 pm	2 hours	Completed integration of staff-related interfaces with the main system and aligned structures to support role-based access. Finalized branch merging with Vidarshan's work and conducted testing using seeded data to ensure system stability prior to the client meeting.
19th March 7:00 am – 8:00 am 9:30 am – 10:30 am 11:00 am – 12:30 pm	3.5 hours	There were lot of debugging and testing prior to the meeting with the client to prepare for the meeting. Once it was done, we were able to successfully demonstrate our app to the client and showed her the progress showing how things were designed and the functionality of the app. Furthermore, taken feedback to improve what we have built. (Meeting with Stephanie and Ghassan)
20th March 10:00 am – 11:00 am	1 hours	Attended project consultation class session to discuss progress, issues, and next steps
21st March 9:00 pm – 10:00 pm 11:00 pm – 1:00 am	3 hours	Improved the admin dashboard (staff related functions only) to provide a smoother and more consistent user experience, with clearer navigation across staff management sections, better visual organization, and a responsive layout that works seamlessly across devices. The corresponding staff PWA functionality sections also redesigned to be more mobile responsive and be wired for the relevant pages.
22nd March 9:00 pm – 10:00 pm 12:30 am – 14:30 pm	3 hours	Went through CRA guidelines and to understand Quebec payroll estimation module, and developed TypeScript

		models covering QPP/QPP2, EI, QPIP, federal tax and Quebec provincial tax.
23rd March 7:00 pm – 8:00 pm 9:00 pm – 11:00 pm	3 hours	Finalized payroll management functionality by completing development and validation of related features, ensuring accurate calculations and stable system behavior. Addressed remaining integration issues and confirmed readiness for deployment. Worked on creating the progress report and updating the AI section and worklogs.
Total	113.75 hours	

3. AI Usage (From 17th March onwards)

Please note that only the principal prompt is included in the table. In certain cases, I had put the entire error message or code, in case I was not able to pinpoint the error/issue. Furthermore, some of these prompts can be longer in length which was needed to debug giving the full context.

AI	Prompt	Value Addition
ChatGPT, Free	How to change my current folder structure to role based access. I have attached my current folder structure image and my partner's setup	To figure out the folder Structure
ChatGPT, Free	What are the good icons for your timesheets and Your pay in react icons io5 I already used these for other items. So let's pick new one. IoCalendarClearOutline, IoCheckboxOutline, IoCloseOutline, IoPersonOutline, IoTimeOutline.	For design improvement for staff related functions in PWA
ChatGPT, Free	Route (app) $\Gamma f / \perp \circ / _not\text{-found} \perp f / \text{admin} \perp f / \text{admin}/\text{clients} \perp f / \text{admin}/\text{clients}/[\text{id}] \perp f / \text{admin}/\text{employees} \perp f / \text{admin}/\text{jobs}/[\text{id}] \perp f / \text{admin}/\text{pay} \perp f / \text{admin}/\text{pay}/[\text{statementId}] \perp f / \text{api}/\text{admin}/\text{pay}\text{-statements} \perp f / \text{api}/\text{ai}/\text{appointments}/[\text{id}]/\text{task}\text{-assistant} \perp f / \text{api}/\text{appointments} \perp f / \text{api}/\text{appointments}/[\text{id}] \perp f / \text{api}/\text{appointments}/[\text{id}]/\text{complete} \perp f / \text{api}/\text{appointments}/[\text{id}]/\text{pause} \perp f / \text{api}/\text{appointments}/[\text{id}]/\text{start} \perp f / \text{api}/\text{appointments}/[\text{id}]/\text{visit}\text{-note} \perp f / \text{api}/\text{assignment} \perp f / \text{api}/\text{auth}/[\text{...nextauth}] \perp f / \text{api}/\text{clients} \perp f / \text{api}/\text{clients}/[\text{id}] \perp f / \text{api}/\text{clients}/[\text{id}]/\text{addresses} \perp f / \text{api}/\text{cron}/\text{appointments} \perp f / \text{api}/\text{jobs} \perp f / \text{api}/\text{jobs}/[\text{id}] \perp f / \text{api}/\text{leave}/[\text{id}] \perp f / \text{api}/\text{mobile}/\text{login} \perp f / \text{api}/\text{staff} \perp f / \text{api}/\text{staff}/[\text{id}] \perp f / \text{api}/\text{staff}/[\text{id}]/\text{availability} \perp f / \text{api}/\text{staff}/[\text{id}]/\text{leave} \perp f / \text{api}/\text{staff}/[\text{id}]/\text{pay}\text{-statements} \perp f / \text{api}/\text{staff}/[\text{id}]/\text{pay}\text{-statements}/[\text{statementId}] \perp f / \text{api}/\text{staff}/[\text{id}]/\text{pay}\text{-statements}/\text{latest} \perp f / \text{api}/\text{staff}/[\text{id}]/\text{time}\text{-sheet} \perp f / \text{api}/\text{staff}/\text{me} \perp f / \text{api}/\text{timesheet}\text{-periods} \perp f / \text{api}/\text{uploadthing} \perp f / \text{api}/\text{uploadthing}/[\text{id}] \perp f / \text{api}/\text{users} \perp f / \text{api}/\text{users}/[\text{id}] \perp \circ / \text{login} \perp \circ / \text{offline} \perp \circ / \text{staff} \perp \circ / \text{staff}/\text{apply}\text{-leave} \perp \circ / \text{staff}/\text{enter}\text{-availability} \perp \circ / \text{staff}/\text{enter}\text{-time} \perp \circ / \text{staff}/\text{pay}\text{-history} \perp \circ / \text{staff}/\text{pay}\text{-periods} \perp \circ / \text{staff}/\text{settings} \perp \circ / \text{staff}/\text{staff}\text{-profile} \perp \circ / \text{staff}/\text{tasks} \perp f / \text{staff}/\text{tasks}/[\text{id}] \perp \circ / \text{staff}/\text{your}\text{-pay}$ What is the meaning, they are not found or offline? How can I fix this?	This is mainly to interpret the messages in ther terminal. This was the first time I ran a full build instead of dev

ChatGPT, Free	How to adjust the spacing between buttons in drawer? same as in HTML? <pre>return (<Container p={0} bg="#f5f6f7" minh="100vh"> <Drawer size="60%" opened={opened} onClose={close} title="Eco Clean"> <Button radius="lg" onClick={() => signOut({ callbackUrl: "/login" })} fullWidth > Staff Profile </Button> <Button radius="lg" onClick={() => signOut({ callbackUrl: "/login" })} fullWidth > Time-off </Button> <Button radius="lg" onClick={() => signOut({ callbackUrl: "/login" })} fullWidth > Availability </Button> <Button radius="lg" color="gray" onClick={() => signOut({ callbackUrl: "/login" })} fullWidth > Logout </Button> </Drawer> </Container>)</pre>	Improving Staff Interfaces
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ChatGPT, Free	import { Stack } from "@mantine/core"; This import did not work, what could be the reason?	Debugging
ChatGPT, Free	What is this error Invalid hook call. Hooks can only be called inside of the body of a function component. This could happen for one of the following reasons: 1. You might have mismatching versions of React and the renderer (such as React DOM) 2. You might be breaking the Rules of Hooks 3. You might have more than one copy of React in the same app See https://react.dev/link/invalid-hook-call for tips about how to debug and fix this problem. <pre>app/(staff)/staff/tasks/page.tsx (62:25) @ module evaluation 60 }; 61 > 62 const router = useRouter(); ^ 63 64 const Page = () => { 65 const { data: session } = useSession();</pre>	Debugging
ChatGPT, Free	What is this syntax error? <pre>Table.Td ta="right"> <Button variant="subtle" size="xs" onClick={router.push('/admin/pay-periods')}>Edit</Button> </Table.Td></pre> How to fix this?	Debugging staff interfaces
ChatGPT, Free	What do you mean by --> NextRouter was not mounted. https://nextjs.org/docs/messages/next-router-not-mounted	Debugging staff interfaces
ChatGPT, Free	I want to generate a shell based or drawer based manage-staff page. In here, I have 3 sections (Text based) so I can click the drawer (collapsible) Leave management Payroll Management Time & Time-Off How can I approach this?	Designing Payroll Related Admin Interface
ChatGPT, Free	How can I keep my original dashboard shell visible when entering a page?	Designing Payroll Related Admin Interface
ChatGPT, Free	eco-clean-web\app\admin\DashboardShell.tsx how to extract the path from this?	To resolve a minor issue when I'm importing the dashboard shell
ChatGPT, Free	How to fix <NavLink label="Payroll Management" component={Link} href="/admin/manage-staff/payroll" leftSection={<IoCashOutline size={18} />} active={{pathname.startsWith("/admin/manage-staff/payroll")}} onClick={() => { setActiveSection("payroll"); close(); }} />	Debugging payroll admin feature pages
ChatGPT, Free	Would you be able to summarize and give the main idea to calculate EI, Tax and all the deductions needs (Federal and in Quebec) based on the PDF	To understand how the deductions are calculated. This was mainly used in design the functions

ChatGPT, Free	<pre> <MenuItem href="/staff/tasks" label="My Tasks" icon={<IoCheckboxOutline size={24} />} active={pathname === '/staff/tasks'} onClick={closeDrawer} /> <MenuItem href="/staff/profile" label="Profile" icon={<IoPersonOutline size={24} />} active={pathname === '/staff/staff-profile'} onClick={closeDrawer} /> <MenuItem href="/staff/time-off" label="Time-off" icon={<IoCalendarClearOutline size={24} />} active={pathname === '/staff/apply-leave'} onClick={closeDrawer} /> <MenuItem href="/staff/availability" label="Availability" icon={<IoTimeOutline size={24} />} active={pathname === '/staff/enter-availability'} onClick={closeDrawer} /> </Stack> why am I not getting routed to each of these places </pre>	Debugging staff interfaces
ChatGPT, Free	<pre> export default function PayHistoryPage() { const { data: session } = useSession(); const [statements, setStatements] = useState<PayStatement[]>([]); const [loading, setLoading] = useState(true); const isMobile = useMediaQuery('(max-width: 768px)'); useEffect(() => { async function fetchHistory() { const userId = '3b32d468-9f20-4808-9f25- bffabed6a9cb'; try { const res = await fetch(`/api/staff/\${userId}/pay- statements`); const data = await res.json(); setStatements(data); } catch (error) { console.error('Error loading history:', error); } finally { setLoading(false); } } how to make the user dynamic by taking the session id ? </pre>	Picking up userId dynamically from the session

ChatGPT, Free	How to fix <code>.next/dev/types/validator.ts:25:44</code> Type error: Type 'Route' does not satisfy the constraint 'LayoutRoutes'. Type 'import("C:/Douglas/AppliedResearch/W26_4495_S2_UpulA/Implementati on/eco-clean-web/.next/dev/types/routes").LayoutRoutes' is not assignable to type 'import("C:/Douglas/AppliedResearch/W26_4495_S2_UpulA/Implementati on/eco-clean-web/.next/types/routes").LayoutRoutes'. Type <code>"/clients"</code> is not assignable to type 'LayoutRoutes'. 23 <code>// This file is generated automatically by Next.js</code> <code>// Do not edit this file manually</code> <code>// This file validates that all pages and layouts export the correct types</code> <code>import type { AppRoutes, LayoutRoutes, ParamMap,</code> <code>AppRouteHandlerRoutes } from "./routes.js"</code> <code>import type { ResolvingMetadata, ResolvingViewport } from "next/types.js"</code> <code>import type { NextRequest } from 'next/server.js'</code> <code>type AppPageConfig<Route extends AppRoutes = AppRoutes> = {</code> <code> default: React.ComponentType< { params: Promise<ParamMap[Route]> } </code> <code>& any> ((props: { params: Promise<ParamMap[Route]> } & any) =></code> <code>React.ReactNode Promise<React.ReactNode> never void </code> <code>Promise<void>)</code> <code> generateStaticParams?: (props: { params: ParamMap[Route] }) =></code> <code>Promise<any[]> any[]</code> <code> generateMetadata?: (</code> <code> props: { params: Promise<ParamMap[Route]> } & any,</code> <code> parent: ResolvingMetadata</code> <code>) => Promise<any> any</code> <code> generateViewport?: (</code> <code> props: { params: Promise<ParamMap[Route]> } & any,</code> <code> parent: ResolvingViewport</code>	Debugging
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```

) => Promise<any> | any
metadata?: any
viewport?: any
}

type LayoutConfig<Route extends LayoutRoutes = LayoutRoutes> = {
  default: React.ComponentType<LayoutProps<Route>> | ((props:
LayoutProps<Route>) => React.ReactNode | Promise<React.ReactNode>
| never | void | Promise<void>)
  generateStaticParams?: (props: { params: ParamMap[Route] }) =>
Promise<any[]> | any[]
  generateMetadata?: (
    props: { params: Promise<ParamMap[Route]> } & any,
    parent: ResolvingMetadata
  ) => Promise<any> | any
  generateViewport?: (
    props: { params: Promise<ParamMap[Route]> } & any,
    parent: ResolvingViewport
  ) => Promise<any> | any
  metadata?: any
  viewport?: any
}

type RouteHandlerConfig<Route extends AppRouteHandlerRoutes =
AppRouteHandlerRoutes> = {
  GET?: (request: NextRequest, context: { params:
Promise<ParamMap[Route]> }) => Promise<Response | void> | Response
| void
  POST?: (request: NextRequest, context: { params:
Promise<ParamMap[Route]> }) => Promise<Response | void> | Response
| void
  PUT?: (request: NextRequest, context: { params:
Promise<ParamMap[Route]> }) => Promise<Response | void> | Response
| void
  PATCH?: (request: NextRequest, context: { params:
Promise<ParamMap[Route]> }) => Promise<Response | void> | Response
| void
  DELETE?: (request: NextRequest, context: { params:
Promise<ParamMap[Route]> }) => Promise<Response | void> | Response
| void

```

```

HEAD?: (request: NextRequest, context: { params:
Promise<ParamMap[Route]> }) => Promise<Response | void> | Response
| void
OPTIONS?: (request: NextRequest, context: { params:
Promise<ParamMap[Route]> }) => Promise<Response | void> | Response
| void
}

// Validate ../../app/(dashboard)/apply-leave/page.tsx
{
  type __IsExpected<Specific extends AppPageConfig<"/apply-leave">> =
Specific
  const handler = {} as typeof import("../../app/(dashboard)/apply-
leave/page.js")
  type __Check = __IsExpected<typeof handler>
  // @ts-ignore
  type __Unused = __Check
}

// Validate ../../app/(dashboard)/clients/[id]/page.tsx
{
  type __IsExpected<Specific extends AppPageConfig<"/clients/[id]">> =
Specific
  const handler = {} as typeof
import("../../app/(dashboard)/clients/[id]/page.js")
  type __Check = __IsExpected<typeof handler>
  // @ts-ignore
  type __Unused = __Check
}

// Validate ../../app/(dashboard)/clients/page.tsx
{
  type __IsExpected<Specific extends AppPageConfig<"/clients">> =
Specific
  const handler = {} as typeof
import("../../app/(dashboard)/clients/page.js")
  type __Check = __IsExpected<typeof handler>
  // @ts-ignore

```

```

    type __Unused = __Check
  }

  // Validate ../../app/(dashboard)/employees/page.tsx
  {
    type __IsExpected<Specific extends AppPageConfig<"/employees">> =
    Specific
    const handler = {} as typeof
import("../app/(dashboard)/employees/page.js")
    type __Check = __IsExpected<typeof handler>
    // @ts-ignore
    type __Unused = __Check
  }

  // Validate ../../app/(dashboard)/enter-availability/page.tsx
  {
    type __IsExpected<Specific extends AppPageConfig<"/enter-
availability">> = Specific
    const handler = {} as typeof import("../app/(dashboard)/enter-
availability/page.js")
    type __Check = __IsExpected<typeof handler>
    // @ts-ignore
    type __Unused = __Check
  }

  // Validate ../../app/(dashboard)/enter-time/page.tsx
  {
    type __IsExpected<Specific extends AppPageConfig<"/enter-time">> =
    Specific
    const handler = {} as typeof import("../app/(dashboard)/enter-
time/page.js")
    type __Check = __IsExpected<typeof handler>
    // @ts-ignore
    type __Unused = __Check
  }

  // Validate ../../app/(dashboard)/jobs/[id]/page.tsx
  {
    type __IsExpected<Specific extends AppPageConfig<"/jobs/[id]">> =

```

<pre> Specific const handler = {} as typeof import("../../app/(dashboard)/jobs/[id]/page.js") type __Check = __IsExpected<typeof handler> // @ts-ignore type __Unused = __Check } // Validate ../../app/(dashboard)/page.tsx { type __IsExpected<Specific extends AppPageConfig<"/">> = Specific const handler = {} as typeof import("../../app/(dashboard)/page.js") type __Check = __IsExpected<typeof handler> // @ts-ignore type __Unused = __Check } // Validate ../../app/(dashboard)/pay-history/page.tsx { type __IsExpected<Specific extends AppPageConfig<"/pay-history">> = Specific const handler = {} as typeof import("../../app/(dashboard)/pay- history/page.js") type __Check = __IsExpected<typeof handler> // @ts-ignore type __Unused = __Check } // Validate ../../app/(dashboard)/pay-periods/page.tsx { type __IsExpected<Specific extends AppPageConfig<"/pay-periods">> = Specific const handler = {} as typeof import("../../app/(dashboard)/pay- periods/page.js") type __Check = __IsExpected<typeof handler> // @ts-ignore type __Unused = __Check } </pre>	
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```

// Validate ../../app/(dashboard)/pay/[statementId]/page.tsx
{
  type __IsExpected<Specific extends
AppPageConfig<"/pay/[statementId]">> = Specific
  const handler = {} as typeof
import("../../app/(dashboard)/pay/[statementId]/page.js")
  type __Check = __IsExpected<typeof handler>
  // @ts-ignore
  type __Unused = __Check
}

// Validate ../../app/(dashboard)/pay/page.tsx
{
  type __IsExpected<Specific extends AppPageConfig<"/pay">> = Specific
  const handler = {} as typeof import("../../app/(dashboard)/pay/page.js")
  type __Check = __IsExpected<typeof handler>
  // @ts-ignore
  type __Unused = __Check
}

// Validate ../../app/(dashboard)/settings/page.tsx
{
  type __IsExpected<Specific extends AppPageConfig<"/settings">> =
Specific
  const handler = {} as typeof
import("../../app/(dashboard)/settings/page.js")
  type __Check = __IsExpected<typeof handler>
  // @ts-ignore
  type __Unused = __Check
}

// Validate ../../app/(dashboard)/staff-profile/page.tsx
{
  type __IsExpected<Specific extends AppPageConfig<"/staff-profile">> =
Specific
  const handler = {} as typeof import("../../app/(dashboard)/staff-
profile/page.js")
  type __Check = __IsExpected<typeof handler>
  // @ts-ignore

```

```

    type __Unused = __Check
  }

  // Validate ../../app/(dashboard)/your-pay/page.tsx
  {
    type __IsExpected<Specific extends AppPageConfig<"/your-pay">> =
    Specific
    const handler = {} as typeof import("../../app/(dashboard)/your-
    pay/page.js")
    type __Check = __IsExpected<typeof handler>
    // @ts-ignore
    type __Unused = __Check
  }

  // Validate ../../app/login/page.tsx
  {
    type __IsExpected<Specific extends AppPageConfig<"/login">> =
    Specific
    const handler = {} as typeof import("../../app/login/page.js")
    type __Check = __IsExpected<typeof handler>
    // @ts-ignore
    type __Unused = __Check
  }

  // Validate ../../app/api/admin/pay-statements/route.ts
  {
    type __IsExpected<Specific extends
    RouteHandlerConfig<"/api/admin/pay-statements">> = Specific
    const handler = {} as typeof import("../../app/api/admin/pay-
    statements/route.js")
    type __Check = __IsExpected<typeof handler>
    // @ts-ignore
    type __Unused = __Check
  }

  // Validate ../../app/api/appointments/[id]/route.ts
  {
    type __IsExpected<Specific extends
    RouteHandlerConfig<"/api/appointments/[id]">> = Specific

```



```

const handler = {} as typeof
import("../../app/api/appointments/[id]/route.js")
type __Check = __IsExpected<typeof handler>
// @ts-ignore
type __Unused = __Check
}

// Validate ../../app/api/appointments/route.ts
{
  type __IsExpected<Specific extends
RouteHandlerConfig<"/api/appointments">> = Specific
  const handler = {} as typeof
import("../../app/api/appointments/route.js")
  type __Check = __IsExpected<typeof handler>
  // @ts-ignore
  type __Unused = __Check
}

// Validate ../../app/api/assignment/route.ts
{
  type __IsExpected<Specific extends
RouteHandlerConfig<"/api/assignment">> = Specific
  const handler = {} as typeof import("../../app/api/assignment/route.js")
  type __Check = __IsExpected<typeof handler>
  // @ts-ignore
  type __Unused = __Check
}

// Validate ../../app/api/auth/[...nextauth]/route.ts
{
  type __IsExpected<Specific extends
RouteHandlerConfig<"/api/auth/[...nextauth]">> = Specific
  const handler = {} as typeof
import("../../app/api/auth/[...nextauth]/route.js")
  type __Check = __IsExpected<typeof handler>
  // @ts-ignore
  type __Unused = __Check
}

```

```

// Validate ../../app/api/clients/[id]/addresses/route.ts
{
  type __IsExpected<Specific extends
RouteHandlerConfig<"/api/clients/[id]/addresses">> = Specific
  const handler = {} as typeof
import("../../app/api/clients/[id]/addresses/route.js")
  type __Check = __IsExpected<typeof handler>
  // @ts-ignore
  type __Unused = __Check
}

// Validate ../../app/api/clients/[id]/route.ts
{
  type __IsExpected<Specific extends
RouteHandlerConfig<"/api/clients/[id]">> = Specific
  const handler = {} as typeof import("../../app/api/clients/[id]/route.js")
  type __Check = __IsExpected<typeof handler>
  // @ts-ignore
  type __Unused = __Check
}

// Validate ../../app/api/clients/route.ts
{
  type __IsExpected<Specific extends
RouteHandlerConfig<"/api/clients">> = Specific
  const handler = {} as typeof import("../../app/api/clients/route.js")
  type __Check = __IsExpected<typeof handler>
  // @ts-ignore
  type __Unused = __Check
}

// Validate ../../app/api/cron/appointments/route.ts
{
  type __IsExpected<Specific extends
RouteHandlerConfig<"/api/cron/appointments">> = Specific
  const handler = {} as typeof
import("../../app/api/cron/appointments/route.js")
  type __Check = __IsExpected<typeof handler>
  // @ts-ignore

```

```

    type __Unused = __Check
  }

  // Validate ../../app/api/jobs/[id]/route.ts
  {
    type __IsExpected<Specific extends
RouteHandlerConfig<"/api/jobs/[id]">> = Specific
    const handler = {} as typeof import("../../app/api/jobs/[id]/route.js")
    type __Check = __IsExpected<typeof handler>
    // @ts-ignore
    type __Unused = __Check
  }

  // Validate ../../app/api/jobs/route.ts
  {
    type __IsExpected<Specific extends RouteHandlerConfig<"/api/jobs">>
= Specific
    const handler = {} as typeof import("../../app/api/jobs/route.js")
    type __Check = __IsExpected<typeof handler>
    // @ts-ignore
    type __Unused = __Check
  }

  // Validate ../../app/api/leave/route.ts
  {
    type __IsExpected<Specific extends RouteHandlerConfig<"/api/leave">>
= Specific
    const handler = {} as typeof import("../../app/api/leave/route.js")
    type __Check = __IsExpected<typeof handler>
    // @ts-ignore
    type __Unused = __Check
  }

  // Validate ../../app/api/mobile/login/route.ts
  {
    type __IsExpected<Specific extends
RouteHandlerConfig<"/api/mobile/login">> = Specific
    const handler = {} as typeof
import("../../app/api/mobile/login/route.js")

```

```

type __Check = __IsExpected<typeof handler>
// @ts-ignore
type __Unused = __Check
}

// Validate ../../app/api/staff/[id]/availability/route.ts
{
  type __IsExpected<Specific extends
RouteHandlerConfig<"/api/staff/[id]/availability">> = Specific
  const handler = {} as typeof
import("../../app/api/staff/[id]/availability/route.js")
  type __Check = __IsExpected<typeof handler>
  // @ts-ignore
  type __Unused = __Check
}

// Validate ../../app/api/staff/[id]/leave/route.ts
{
  type __IsExpected<Specific extends
RouteHandlerConfig<"/api/staff/[id]/leave">> = Specific
  const handler = {} as typeof
import("../../app/api/staff/[id]/leave/route.js")
  type __Check = __IsExpected<typeof handler>
  // @ts-ignore
  type __Unused = __Check
}

// Validate ../../app/api/staff/[id]/pay-statements/[statementId]/route.ts
{
  type __IsExpected<Specific extends
RouteHandlerConfig<"/api/staff/[id]/pay-statements/[statementId]">> =
Specific
  const handler = {} as typeof import("../../app/api/staff/[id]/pay-
statements/[statementId]/route.js")
  type __Check = __IsExpected<typeof handler>
  // @ts-ignore
  type __Unused = __Check
}

```

```

// Validate ../../app/api/staff/[id]/pay-statements/latest/route.ts
{
  type __IsExpected<Specific extends
RouteHandlerConfig<"/api/staff/[id]/pay-statements/latest">> = Specific
  const handler = {} as typeof import("../../app/api/staff/[id]/pay-
statements/latest/route.js")
  type __Check = __IsExpected<typeof handler>
  // @ts-ignore
  type __Unused = __Check
}

// Validate ../../app/api/staff/[id]/pay-statements/route.ts
{
  type __IsExpected<Specific extends
RouteHandlerConfig<"/api/staff/[id]/pay-statements">> = Specific
  const handler = {} as typeof import("../../app/api/staff/[id]/pay-
statements/route.js")
  type __Check = __IsExpected<typeof handler>
  // @ts-ignore
  type __Unused = __Check
}

// Validate ../../app/api/staff/[id]/route.ts
{
  type __IsExpected<Specific extends
RouteHandlerConfig<"/api/staff/[id]">> = Specific
  const handler = {} as typeof import("../../app/api/staff/[id]/route.js")
  type __Check = __IsExpected<typeof handler>
  // @ts-ignore
  type __Unused = __Check
}

// Validate ../../app/api/staff/me/route.ts
{
  type __IsExpected<Specific extends
RouteHandlerConfig<"/api/staff/me">> = Specific
  const handler = {} as typeof import("../../app/api/staff/me/route.js")
  type __Check = __IsExpected<typeof handler>
  // @ts-ignore

```

```

    type __Unused = __Check
  }

  // Validate ../../app/api/staff/route.ts
  {
    type __IsExpected<Specific extends RouteHandlerConfig<"/api/staff">>
    = Specific
    const handler = {} as typeof import("../../app/api/staff/route.js")
    type __Check = __IsExpected<typeof handler>
    // @ts-ignore
    type __Unused = __Check
  }

  // Validate ../../app/api/staff/time-sheet/route.ts
  {
    type __IsExpected<Specific extends
RouteHandlerConfig<"/api/staff/time-sheet">> = Specific
    const handler = {} as typeof import("../../app/api/staff/time-
sheet/route.js")
    type __Check = __IsExpected<typeof handler>
    // @ts-ignore
    type __Unused = __Check
  }

  // Validate ../../app/api/timesheet-entry/route.ts
  {
    type __IsExpected<Specific extends RouteHandlerConfig<"/api/timesheet-
entry">> = Specific
    const handler = {} as typeof import("../../app/api/timesheet-
entry/route.js")
    type __Check = __IsExpected<typeof handler>
    // @ts-ignore
    type __Unused = __Check
  }

  // Validate ../../app/api/timesheet-periods/route.ts
  {
    type __IsExpected<Specific extends RouteHandlerConfig<"/api/timesheet-
periods">> = Specific

```

```

const handler = {} as typeof import("../../app/api/timesheet-
periods/route.js")
type __Check = __IsExpected<typeof handler>
// @ts-ignore
type __Unused = __Check
}

// Validate ../../app/api/uploadthing/[id]/route.ts
{
  type __IsExpected<Specific extends
RouteHandlerConfig<"/api/uploadthing/[id]">> = Specific
  const handler = {} as typeof
import("../../app/api/uploadthing/[id]/route.js")
  type __Check = __IsExpected<typeof handler>
  // @ts-ignore
  type __Unused = __Check
}

// Validate ../../app/api/uploadthing/route.ts
{
  type __IsExpected<Specific extends
RouteHandlerConfig<"/api/uploadthing">> = Specific
  const handler = {} as typeof import("../../app/api/uploadthing/route.js")
  type __Check = __IsExpected<typeof handler>
  // @ts-ignore
  type __Unused = __Check
}

// Validate ../../app/api/users/[id]/route.ts
{
  type __IsExpected<Specific extends
RouteHandlerConfig<"/api/users/[id]">> = Specific
  const handler = {} as typeof import("../../app/api/users/[id]/route.js")
  type __Check = __IsExpected<typeof handler>
  // @ts-ignore
  type __Unused = __Check
}

// Validate ../../app/api/users/route.ts

```

```

{
  type __IsExpected<Specific extends RouteHandlerConfig<"/api/users">>
= Specific
  const handler = {} as typeof import("../..../app/api/users/route.js")
  type __Check = __IsExpected<typeof handler>
  // @ts-ignore
  type __Unused = __Check
}

// Validate ../..../app/(dashboard)/clients/layout.tsx
{
  type __IsExpected<Specific extends LayoutConfig<"/clients">> = Specific
  const handler = {} as typeof
import("../..../app/(dashboard)/clients/layout.js")
  type __Check = __IsExpected<typeof handler>
  // @ts-ignore
  type __Unused = __Check
}

// Validate ../..../app/(dashboard)/employees/layout.tsx
{
  type __IsExpected<Specific extends LayoutConfig<"/employees">> =
Specific
  const handler = {} as typeof
import("../..../app/(dashboard)/employees/layout.js")
  type __Check = __IsExpected<typeof handler>
  // @ts-ignore
  type __Unused = __Check
}

// Validate ../..../app/(dashboard)/jobs/[id]/layout.tsx
{
  type __IsExpected<Specific extends LayoutConfig<"/jobs/[id]">> =
Specific
  const handler = {} as typeof

```



```

import("../../app/(dashboard)/jobs/[id]/layout.js")
type __Check = __IsExpected<typeof handler>
// @ts-ignore
type __Unused = __Check
}

// Validate ../../app/(dashboard)/layout.tsx
{
  type __IsExpected<Specific extends LayoutConfig<"/">> = Specific
  const handler = {} as typeof import("../../app/(dashboard)/layout.js")
  type __Check = __IsExpected<typeof handler>
  // @ts-ignore
  type __Unused = __Check
}

// Validate ../../app/layout.tsx
{
  type __IsExpected<Specific extends LayoutConfig<"/">> = Specific
  const handler = {} as typeof import("../../app/layout.js")
  type __Check = __IsExpected<typeof handler>
  // @ts-ignore
  type __Unused = __Check
}

24 | type LayoutConfig<Route extends LayoutRoutes = LayoutRoutes> = {
> 25 |   default: React.ComponentType<LayoutProps<Route>> | ((props:
LayoutProps<Route>) => React.ReactNode | Promise<React.ReactNode>
| never | void | Promise<void>)
    |
    ^
26 |   generateStaticParams?: (props: { params: ParamMap[Route] }) =>
Promise<any[]> | any[]
27 |   generateMetadata?: (
28 |     props: { params: Promise<ParamMap[Route]> } & any,
Next.js build worker exited with code: 1 and signal: null

```

Thought for 24s

This is usually not a problem in validator.ts itself. That file is generated by Next.js, so the real issue is almost always one of these:

- stale generated route types in .next
- a layout whose props don't match the route
- a parallel route / slot changing the expected layout shape
- a current typed-routes generation bug/mismatch between dev and build output

Next.js generates route-aware helpers like LayoutProps during next dev, next build, or next typegen, and layouts are expected to match those generated route types.

What your error is really saying

Your generated validator is comparing:

.next/dev/types/routes
against .next/types/routes

and they disagree about what LayoutRoutes contains. In your error, "/clients" exists in one generated type set but not the other. That strongly suggests a generated-types mismatch, not a handwritten TypeScript mistake.

Fix in this order

1) Stop the dev server and delete .next

This is the first thing I would do.

```
# stop next dev first  
rmdir /s /q .next
```

Then run:

```
npm run build
```

Or if you want to regenerate types first:

```
npx next typegen
```

npm run build

Because Next.js generates these route-aware types during dev/build/typegen, clearing .next often fixes route-type drift.

2) Check app/(dashboard)/clients/layout.tsx

Your validator shows this exact route is being checked:

type __IsExpected<Specific extends LayoutConfig<"/clients">> = Specific

So app/(dashboard)/clients/layout.tsx must be a valid layout for /clients.

Use one of these two safe forms.

Safe simple version

```
import type { ReactNode } from 'react';
```

```
export default function ClientsLayout({  
  children,  
}: {  
  children: ReactNode;  
}) {  
  return <>{children}</>;  
}
```

Or the route-aware helper version

```
export default function ClientsLayout(props: LayoutProps<"/clients">) {  
  return <>{props.children}</>;  
}
```

The official docs say LayoutProps is the generated helper for layouts.

3) Check for accidental extra props in that layout

A layout for /clients should normally only need:

children

and sometimes params for dynamic segments

If your clients/layout.tsx is something like this:

```
export default function Layout({
  children,
  client,
  params,
}): {
  children: React.ReactNode;
  client: any;
  params: any;
}) {
```

that can break typing unless those props come from actual Next.js route conventions.

For a non-dynamic route like /clients, do not invent props.

4) Check for any parallel route folders like @something

If there is a folder such as:

app/(dashboard)/clients/@sidebar

then the layout props change, and Next expects an extra slot prop like sidebar. The official layout typing includes slot props for parallel routes, and Next maintainers have also pointed out that parallel route folders change what LayoutProps requires.

So if you have something like:

app/(dashboard)/clients/@details

your layout may need:

```
export default function ClientsLayout({
  children,
  details,
}): LayoutProps<'/clients'> {
  return (
```

```
<>
  {details}
  {children}
</>
);
}
```

If you do not intend to use parallel routes, remove the @... folder.

5) Make sure you are not running dev and build against stale output

Because your error references both:

```
.next/dev/types/...
.next/types/...
```

it can happen when route generation changed during dev and build is validating against a different snapshot. There is also a recent reported Next.js issue around typed routes and route generation mismatches in certain setups.

So use this clean cycle:

```
# stop dev server
rmdir /s /q .next
npm install
npm run build
```

Then restart dev only after build succeeds.

6) Keep clients/layout.tsx minimal

For now, make it as small as possible:

```
import type { ReactNode } from 'react';

export default function Layout({
  children,
}): {
  children: ReactNode;
```

```
  }) {  
    return <> {children}</>;  
  }  
}
```

If this builds, then the problem was your layout props.

Most likely cause in your case

Because the error specifically says:

"/clients" is not assignable to type LayoutRoutes

and it compares .next/dev/types/routes to .next/types/routes,

the most likely cause is:

you changed route/layout structure recently,
.next kept stale generated route types,
and clients/layout.tsx is now being validated against conflicting generated
route maps.

Best practical fix

Do this exactly:

1. stop next dev

2. delete build cache

rmdir /s /q .next

3. optional but helpful

npx next typegen

4. build again

npm run build

Then make sure app/(dashboard)/clients/layout.tsx is:

```
import type { ReactNode } from 'react';
```

```
export default function Layout({
```

```
children,  
}: {  
  children: ReactNode;  
}) {  
  return <>{children}</>;  
}
```

And check that there is no accidental @slot folder under clients.

One more thing

Do not edit:

`.next/dev/types/validator.ts`

That file is generated automatically by Next.js. The docs explicitly note these route-aware layout helpers are generated during dev/build/typegen.

Paste your `app/(dashboard)/clients/layout.tsx` here and I'll point out the exact line that is causing it.

I changed the folder structure as follows:

Thought for 14s

Yes — this folder change explains the error.

You changed from routes like:

`app/(dashboard)/clients/page.tsx -> /clients`
`app/(dashboard)/employees/page.tsx -> /employees`

to something like:

`app/(admin)/admin/clients/page.tsx -> /admin/clients`
`app/(admin)/admin/employees/page.tsx -> /admin/employees`

`app/(staff)/staff/apply-leave/page.tsx -> /staff/apply-leave`

In Next.js, route groups like (admin) and (staff) do not appear in the URL, but normal folders like admin and staff do appear in the URL. So your new structure changed the actual route paths.

That means the old generated type checker was still expecting routes such as `"/clients"` and `"/employees"`, while your new app structure now produces `"/admin/clients"` and `"/admin/employees"`. The route-aware helpers and validator are generated into `.next`, so after a route-structure change, stale generated files can cause exactly this kind of mismatch.

What to do

Option 1: If you want URLs like `/admin/clients` and `/staff/apply-leave`

Your current structure is conceptually fine. Then you should:

stop the dev server
delete `.next`
rebuild

On Windows:

`rmdir /s /q .next`
`npm run build`

You can also regenerate route types first:

`npx next typegen`
`npm run build`

Next.js generates these route-aware types during `next dev`, `next build`, or `next typegen`, so clearing `.next` after a folder restructure is the right first fix.

Option 2: If you still want URLs like `/clients` and `/employees`

Then your new structure is wrong for that goal.

Use:

app
 (admin)
 clients
 page.tsx
 employees
 page.tsx
 jobs
 page.tsx
 pay
 page.tsx
 layout.tsx
 page.tsx

not:

app
 (admin)
 admin
 clients
 employees

Because admin is a real route segment, it becomes part of the URL. Route groups in parentheses are only for organization and are omitted from the path.

So:

app/(admin)/clients/page.tsx → /clients
app/(admin)/admin/clients/page.tsx → /admin/clients

Why your exact error happened

Your validator still showed checks like:

LayoutConfig<"/clients">

That means the generated route map still had the old route shape in one place, while another generated route map had the new one. The mismatch between .next/dev/types/routes and .next/types/routes is a generated-types inconsistency, not something you should fix inside validator.ts. There is also a recently reported issue around generated validator/type imports in

.next/dev/types, so clearing and rebuilding is especially important here.

Best fix for your project right now

If your intention is:

admin area under /admin/...

staff area under /staff/...

then keep the structure, but clean the generated output:

```
rmdir /s /q .next
```

```
npx next typegen
```

```
npm run build
```

Also check your layouts:

```
app/(admin)/admin/layout.tsx
```

```
app/(staff)/staff/layout.tsx
```

They should simply accept children unless you intentionally use more advanced route features. Layouts are tied to route segments, so after moving folders, the layout location must match the route tree.

A safe version is:

```
import type { ReactNode } from 'react';
```

```
export default function Layout({ children }: { children: ReactNode }) {  
  return <>{children}</>;  
}
```

My guess

Your real issue is this sequence:

old routes were /clients, /employees, etc.

you moved them under /admin/... and /staff/...

.next still had old route type output

build compared old generated types with new generated types and failed

So the first thing I would do is:

```
rmdir /s /q .next  
npm run build
```

If it still fails after that, the next thing to inspect is whether any file still imports or references old route strings like `"/clients"` directly.

Paste your new app tree in text form, or paste the two layout files under admin and staff, and I'll tell you exactly whether the structure itself is correct.

what does `rmdir /s /q .next` do?

remove folders of what?

Good question — this is important to understand 🧐

What `rmdir /s /q .next` does

It is a Windows command that deletes the `.next` folder completely.

Breakdown:

- `rmdir` → remove directory
- `/s` → delete all subfolders and files inside
- `/q` → quiet mode (no confirmation prompts)
- `.next` → the folder being deleted

So this command means:

🔪 “Force delete the entire `.next` folder and everything inside it.”

What is the `.next` folder?

This folder is automatically created by Next.js.

It contains:

1. Build output

Compiled pages
Server bundles
Static assets
2. Cache
Faster rebuilds
Cached modules
3. Generated types (important for your error)
types/routes.ts
dev/types/validator.ts
route-aware typing

🔗 This is exactly where your error is coming from.

Why deleting .next fixes your issue

When you changed your folder structure:

/clients → /admin/clients

Next.js still had old cached route types like:

"/clients"

But your new app now generates:

"/admin/clients"

So:

old types ≠ new types ✖
build fails

Deleting .next forces Next.js to:

- ☒ regenerate everything from scratch
- ☒ rebuild correct route types
- ☒ remove stale cached files

Important: Is it safe?

✓ 100% safe

You are NOT deleting your code

You are NOT deleting your database

You are NOT deleting node_modules

You are only deleting:

👉 temporary build/cache files

Next.js will recreate .next automatically when you run:

npm run dev

or

npm run build

Equivalent commands (just so you know)

Windows (you used this)

rmdir /s /q .next

Mac/Linux

rm -rf .next

When should you delete .next?

Do it when:

You change folder structure (like you did)

You rename routes/layouts

Strange TypeScript errors from .next/types

Build works weirdly but code looks correct

Short version

👉 .next = Next.js cache + build + generated types

👉 Deleting it = reset Next.js state

If you want, I can also show you how your new /admin + /staff structure maps to actual URLs (this is where most people get confused).

PS

C:\Douglas\AppliedResearch\W26_4495_S2_UpulA\Implementation\eco-

```
clean-web> rmdir /s /q .next
Remove-Item : A positional parameter cannot be found that accepts argument
'/q'.
At line:1 char:1
+ rmdir /s /q .next
+ ~~~~~~
+ CategoryInfo          : InvalidArgument: (:) [Remove-Item],
ParameterBindingException
+ FullyQualifiedErrorId :
PositionalParameterNotFound,Microsoft.PowerShell.Commands.RemoveItemCommand
```

Ahh 🤖 — you're using PowerShell, not Command Prompt. That's why it failed.

Why the error happened

In PowerShell:

rmdir is actually an alias for Remove-Item
It does NOT support /s /q flags (those are for Command Prompt)

So PowerShell is saying:

“I don't understand /q”

☒ Correct command for PowerShell

Use this instead:

```
Remove-Item -Recurse -Force .next
```

What it means:

-Recurse → delete all files inside (like /s)

-Force → no prompts + delete hidden files (like /q)



Steps you should follow now

1. Stop your dev server (Ctrl + C)

2. Delete .next

```
Remove-Item -Recurse -Force .next
```

3. Rebuild
npm run build

⚡ Alternative (if you want Command Prompt style)

You can also run:

```
cmd /c "rmdir /s /q .next"
```

But honestly, just use:

```
Remove-Item -Recurse -Force .next
```

🔍 If you get "access denied"

Sometimes files are locked. Then do:

```
Remove-Item -Recurse -Force .next -ErrorAction SilentlyContinue
```

☑ After this

Your error should go away because:

old /clients route types → deleted

new /admin/clients routes → regenerated cleanly

If it still fails after this, next step is:

👉 I'll check your layout.tsx files (very likely second cause)

Just tell me 🙋

Failed to compile.

```
./lib/ai/index.ts
```

```
Module not found: Can't resolve 'openai'
```

<https://nextjs.org/docs/messages/module-not-found>

Import trace for requested module:

```
./app/api/ai/appointments/[id]/task-assistant/route.ts
```

```
./lib/ai/schemas/index.ts
Module not found: Can't resolve 'zod'
```

<https://nextjs.org/docs/messages/module-not-found>

Import trace for requested module:
./app/api/ai/appointments/[id]/task-assistant/route.ts

> Build failed because of webpack errors

What are webpack errors?

Good — now you've moved past the route issue 👍
This new error is much simpler.

🔍 What your error means
Module not found: Can't resolve 'openai'
Module not found: Can't resolve 'zod'

👉 Your code is trying to use:

```
openai
zod
```

But they are NOT installed in your project

🤔 What are Webpack errors?
Simple explanation (your style)

👉 Webpack = bundler

It takes all your code:

```
import { something } from "some-library";
```

and tries to:

Find that library
Combine everything into a build
Prepare it to run
So a Webpack error means:

👉 “I tried to build your app, but something is missing or broken”

In your case:

✗ Webpack cannot find:

openai
zod

So it fails the build.

★ Why this happened?

ChatGPT, Free	Why my command is not working and giving this error? PS C:\Douglas\AppliedResearch\W26_4495_S2_UpulA\Implementation\eco-clean-web> rmdir /s /q .next Remove-Item : A positional parameter cannot be found that accepts argument '/q'. At line:1 char:1 + rmdir /s /q .next + ~~~~~~ + CategoryInfo : InvalidArgument: (:) [Remove-Item], ParameterBindingException + FullyQualifiedErrorId : PositionalParameterNotFound,Microsoft.PowerShell.Commands.RemoveItemCommand	Debugging
ChatGPT, Free	Failed to compile. ./lib/ai/index.ts Module not found: Can't resolve 'openai' https://nextjs.org/docs/messages/module-not-found Import trace for requested module: ./app/api/ai/appointments/[id]/task-assistant/route.ts ./lib/ai/schemas/index.ts Module not found: Can't resolve 'zod' https://nextjs.org/docs/messages/module-not-found Import trace for requested module: ./app/api/ai/appointments/[id]/task-assistant/route.ts > Build failed because of webpack errors What are webpack errors?	After bringing the main branch changes to my staff branch, had to fix some compilation errors
ChatGPT, Free	Property 'leave' does not exist on type 'PrismaClient<PrismaClientOptions, never, DefaultArgs>'.ts(2339) But, leave table is there	After bringing the main branch changes to my staff branch, some functions did not work. Had to debug
ChatGPT, Free	import type { NextRequest } from 'next/server.js'; import type { ResolvingMetadata, ResolvingViewport } from 'next/types.js'; import type { AppRouteHandlerRoutes, AppRoutes, LayoutRoutes, ParamMap, } from './routes.js'; Cannot find module './routes.js' or its corresponding type declarations.ts(2307)	Debugging
ChatGPT, Free	how to find out my postgres local db, can I reset password?	Local DB password issue
ChatGPT, Free	"seed": "tsx prisma/seed.ts", "seed:small": "tsx prisma/seed.ts small", "seed:medium": "tsx prisma/seed.ts medium", "seed:large": "tsx prisma/seed.ts large" }, do I have to add tiny in here?	This was done as a part of modifying Vidarshan's work on seeding data to match the full db schema

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